Lie Groups — Lecture 3

November 7, 2026

1 Reminder

Definition 1.1 (Tangent space)**.** If (X, A) is a smooth manifold and $p \in X$ is some point, then the tangent space of p is defined as,

$$T_pX = \{\gamma : (-\delta, \delta) \to X \mid \gamma(0) = p, \gamma \in C^\infty(X), \delta \in \mathbb{R}^+\}/\sim$$

The tangent bundle of X is denoted $TX = \bigsqcup_{p\in X} T_pX$.

Definition 1.2 (exponential map)**.** If G is a Lie group then define $\exp : L(G) \to G$ as the map that for any $v \in L(G)$, θ_v be the corresponding 1-parameter subgroup of G, then $\exp tv = \theta_v(t)$.

Proposition 1.3. *Let $\varphi_e : U_e \to \Omega_0$ be a chart where $e \in U_e \subseteq G$ and $0 \in \Omega_0 \subseteq L(G)$ open neighborhoods. Let $x, y \in \Omega_0$ and denote $x \cdot y = \varphi(\varphi^{-1}(x) \cdot \varphi^{-1}(y))$, then $x \cdot y = x + y + o(r)$ where r is the distance of (x, y) and (e, e).*

2 Submanifolds

We intend to characterize the subspace of a Lie group, in order to do so we will discuss a subspace retaining the properties of manifold, as the characterization of a subgroup is well defined. The general idea if to define submanifold as the restriction of a manifold to a subspace such that its smooth property, as show up as smoothness of charts, would not be affected.

Definition 2.1 (Submanifold)**.** Let M be a smooth manifold modeled on V, and let $N \subseteq M$ be a subset and $W \subseteq V$ be a subspace. N is a **submanifold** modeled on W if for any $x \in N$ there exists a chart compatible with M, $\varphi_\alpha : M_\alpha \to \Omega_\alpha$ where $M_\alpha \subseteq M, \Omega_\alpha \subseteq V$ are open, that restricts to a bijection $M_\alpha \cap N \to \Omega_\alpha \cap W$.

Remark. The requirement for a compatible chart is crucial, as the given atlas might not contain a single chart matching M_α, instead being equivalent to an atlas that does.

Our first task is to show that a submanifold, in the sense of a manifold subset retaining smoothness properties, is indeed a smooth manifold on its own. We first need an actual atlas to show that, that leads us to,

Proposition 2.2. *Given a manifold M and a submanifold N, the bijections $\{M_\alpha \cap N \to \Omega_\alpha \cap W\}$ are an atlas on N. Furthermore, equivalent atlases on M yield equivalent atlases on N thus N is a smooth manifold modeled on W.*

Proof. Left as an exercise. □

We can now describe the structure of a submanifold as a manifold,

Proposition 2.3. *If N is a submanifold of M, then N can be given differential structure as a manifold so that:*

1. *The inclusion of N in M is smooth*
2. *$P \to N$ is smooth if and only if the composition $P \to N \to M$ is smooth*

Proof. Let $f : N \to M$ be the identity embedding, and let $\varphi : U \to V$ be a chart of M, where $U \subseteq M, V \subseteq \mathbb{R}^k$ are open, and assume $N \cap U \neq \emptyset$. Then φ can be restricted to $\psi : N \cap U \to V \cap W$ a bijection and $\varphi \circ f \circ \psi$ is smooth by our definition of smoothness and as a restriction.

The second part is similar and is left as an exercise. □

Corollary 2.4. $T(N)$ is embedded in $T(M)$.

Recall that if M_1, M_2 are smooth manifolds, $M_1 \times M_2$ can be given a smooth manifold structure, this holds for submanifolds as well.

Exercise 2.1. Show that if N_1, N_2 are submanifolds of M_1, M_2 then $N_1 \times N_2$ is a submanifold of $M_1 \times M_2$, and its differential structure as a submanifold is the same as of the product of the submanifolds as smooth manifolds.

As of now, there are two distinct "substructures" we consider of a Lie group, that is the subgroup & the submanifold. The question that naturally arises is whether there is a connection between these two substructures.

Proposition 2.5. *If G is a Lie group and H is both a submanifold and a subgroup then H assumes the structure of a Lie group.*

Proof. Follows from the last exercise and proposition for the maps $\mu : G \times G \to G$ and $\mu : H \times H \to H$ and $i : G \to G$ to $i : H \to H$. To elaborate we defined the submanifold differential structure as such that preserves smooth maps, then the group operations (that are of course smooth) are being preserved. □

So these two structure do collide when in Lie groups.

3 Closed subgroups

While subgroups are a natural way to understand groups, and are easy to construct, submanifolds are somewhat more difficult object to handle, is there a way to construct submanifolds from subgroups? We will show that in a special case of subgroups the answer is that it is.

Definition 3.1 (Closed subgroup)**.** A subgroup H of a Lie group G is closed if it is closed as a subset of the topology G.

To understand the case of a non-closed subgroup:

Example 3.1. Let $M = T^2 = \mathbb{R}^2/\mathbb{Z}^2$ the 2-torus, and take $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ some irrational number. Define $H = \{(t, \alpha t) \in \mathbb{R}^2 \mid t \in \mathbb{R}\}/\mathbb{Z}^2$, then $H \leq M$ a subgroup, and a dense set of $\mathbb{R}^2/\mathbb{Z}^2$, in particular it isn't closed.

It is also clear that H does not assume a manifold structure, as any small enough open neighborhood of any point $\in H$ is not connected as $H \cap (\mathbb{Q}/\mathbb{Z})^2 = \{0\}$.

In the other hand,

Example 3.2. If $\alpha \in \mathbb{Q}$ then there must be $z \in \mathbb{Z} \setminus \{0\}$ such that $\alpha z \in \mathbb{Z}$, meaning that H is a closed path in the topological space M. In particular H is indeed a 1 dimensional manifold.

This result is no coincidence, it leads us to,

Theorem 3.2 (Closed subgroup as submanifold)**.** *A closed subgroup H of a Lie group G is a submanifold as well.*

The proof of the theorem is given by the next three lemmas:

Lemma 3.3. *Suppose $L(G) = T_eG$ has a norm $\|\cdot\|$, and suppose that $\{h_n\}_{n=1}^{\infty} \subseteq L(G) \setminus \{0\}$ is a sequence of points such that,*

$$\forall n \in \mathbb{N},\ \exp h_n \in H, \qquad h_n \xrightarrow{n\to\infty} 0, \qquad \frac{1}{\|h_n\|} h_n \to v \in L(G)$$

Then $\exp(tv) \in H$ *for all* $t \in \mathbb{R}$.

In other words, a closed subgroup is closed under "directions", namely if a direction vectors converge then the span of the result is in W.

Proof. Suppose $t \in \mathbb{R}$, then,

$$\frac{t}{\|h_n\|} h_n \to tv$$

and $\|h_n\| \to 0$. Let $\{m_n\} \subseteq \mathbb{N}$ be a sequence such that $m_n\|h_n\| \xrightarrow{n\to\infty} t$, such sequence can be obtained directly by limit definition. Then,

$$\lim_{n\to\infty} \exp m_n h_n = \lim_{n\to\infty} \exp \frac{m_n\|h_n\|}{\|h_n\|} h_n = \lim_{n\to\infty} \exp m_n\|h_n\| \cdot \lim_{n\to\infty} \exp \frac{1}{\|h_n\|} h_n = \exp tv$$

But $\exp m_n h_n = (\exp h_n)^{m_n} \in H$, it follows from H's closure that $\exp tv \in H$. □

Lemma 3.4. *Let $W \subseteq L(G)$ be a set of all tv for $t \in \mathbb{R}$ and $v \in L(G)$ of the last lemma. Then $\exp W \subseteq H$ and W is a vector subspace of $L(G)$.*

Proof. If $v \in L(G)$ as in the last lemma, then $\exp tv \in H$, W consists of such values therefore $\exp W \subseteq H$.

It is clear that if $w \in W$ then $tw \in W$ for all $t \in \mathbb{R}$, and it remain to show that if $w_1, w_2 \in W$ then $w_1 + w_2 \in W$ as well. The case $w_1 + w_2 = 0$ is clear then assume otherwise. Consider $g(t) = \exp(tw_1)\exp(tw_2)$ and recall that exp is smooth, $g(0) = \exp(0)\exp(0) = e$, then for small enough values of t, there is a smooth map $g(t) = \exp f(t)$ with $f(0) = 0$ and $f(t) \in L(G)$.

Recall that $x \cdot y = x + y + o(r)$ for distance r in $G \times G$, when $x, y \in$ a small enough neighborhood of e, then the same holds for f,

$$g(t) = \exp f(t) = \exp(tw_1 + tw_2) + o(t)$$

then $\frac{1}{t} f(t) \xrightarrow{t\to 0} w_1 + w_2$, by choosing $h_n = f(\frac{1}{n})$ the conditions for the last lemma hold for $v = \frac{w_1+w_2}{\|w_1+w_2\|}$, therefore $w_1 + w_2 \in W$. □

Lemma 3.5. *$\exp W$ is a neighborhood of $e \in H$.*

Proof. Split $L(G) = W' \oplus W$, there is a diffeomorphism $\varphi(w', w) = \exp(w')\exp(w)$ between a neighborhood of $0 \in L(G)$ and a neighborhood of $e \in G$. Assume the lemma does not hold for contradiction, then there is no open set $e \in \exp U \subseteq \exp W \subseteq H$. In other words there is a sequence $\{h_n\} \subseteq H$ such that $h_n \to e$. Denote $h_n = \varphi(w'_n, w_n)$, then by the contradiction it can be assumed that $w'_n \neq 0$ for all n. Since H is a subgroup and $\exp w_n \in H, \varphi(w'_n, w_n) \in H$ it follows that $\exp w'_n \in H$ as well. $\frac{w'_n}{\|w'_n\|} \in S^k$ for $S^k = \{v \in L(G) \mid \|v\| = 1\}$ a k-sphere, in particular in a compact set, then there is a subsequence n_i such that $\frac{w'_{n_i}}{\|w'_{n_i}\|} \to w'$ for some $w' \in W' \cap S^k$. It follows from the first lemma that $w' \in W$, then $w' \in W \cap W' \implies w' = 0$ which is contradiction to $\|w'\| = 1$. □

Proof of 3.2. By the last lemma $\exp W$ is a neighborhood of e in H, thus $\exp|_W$ is a chart of a neighborhood of e. This chart can be translated to any other point $\in H$ by left multiplication, thus generating an atlas to H, inducing a smooth manifold structure. □

Example 3.3. Consider $O(n) \subseteq \mathrm{GL}_n(\mathbb{R})$ the subgroup of orthogonal invertible matrices over the reals. Define $f : \mathrm{GL}_n(\mathbb{R}) \to \mathrm{GL}_n(\mathbb{R})$ by $f(A) = A^T A$ then f is continuous and $O(n) = f^{-1}(\{\mathrm{id}\})$. Then $O(n)$ is closed and therefore by 3.2 a submanifold and a Lie group.

Example 3.4. $U(n) \subseteq \mathrm{GL}_n(\mathbb{C})$ the subspace of unitary complex matrices is also a closed subset as a similar case of $O(n)$, thus also a Lie group.

As an embedded manifold we can identify $T_{\mathrm{id}}(O(n))$ and $T_{\mathrm{id}}(U(n))$ as subspaces of $M_n(\mathbb{R})$ and $M_n(\mathbb{C})$.

Proposition 3.6. *In both Lie groups, $L(G) = \{X \in M_n(\mathbb{F}) \mid \bar{X}^T = -X\}$.*

Proof. If $X \in L(G)$ then there is a smooth curve $\gamma(t) = I_n + tX + o(t)$, and $\gamma(t) \in O(n)$ for all t,

$$\overline{f(t)}^T f(t) = I$$

equivalently,

$$(I + t\bar{X}^T + o(t))(I + tX + o(t)) = I \implies \bar{X}^T + X = 0$$

From the other hand, if $\bar{X}^T = -X$ then,

$$\overline{(\exp tX)}^T = \overline{\left(\sum_{n=0}^{\infty} \frac{t^n X^n}{n!}\right)}^T = \sum_{n=0}^{\infty} \frac{t^n (\bar{X}^T)^n}{n!} = \sum_{n=0}^{\infty} \frac{t^n (-X)^n}{n!} = (\exp tX)^{-1}$$

which means that $\exp tX$ is in the subgroup for each case, then $X \in L(G)$. □

Example 3.5. If G is a compact Lie group and H is a closed connected Abelian subgroup, then H is a torus.

Proposition 3.7. *A closed connected subgroup H of a Lie group G is determined by its tangent space at e.*

Proof. We saw that a homomorphism $G \to H$ of Lie groups is determined by the induced homomorphism $L(G) \to L(H)$ when G is connected. Combining this proposition together with the result that the tangent space is a subspace of $L(G)$ gives us full characterization by tangent space. □

4 Quotient spaces

Definition 4.1 (Subgroup & Cosets)**.** Suppose G is a Lie group and H is a closed subgroup, then the **quotient space** G/H is the set of (left) cosets $\{gH \mid g \in G\}$. The map $\pi : G \to G/H$ given by $g \mapsto gH$ is the projection, and give G/H the quotient topology.

Proposition 4.2. *G/H is Hausdorff.*

Proof. the projection $p : G \to G/H$ is an open map, and define $R = \{(x, y) \in G \times G \mid x^{-1}y \in H\}$, meaning that $(x, y) \in R \iff xH = yH$. Define $f : G \times G \to G$ by $f(x, y) = x^{-1}y$, f is continuous as both the group multiplication & inverse are continuous, and $f^{-1}(H) = R$. $H \subseteq G$ is closed and f is continuous therefore $f^{-1}(H) = R$ is closed in $G \times G$.

By the property that a topological quotient space X/H is Hausdorff iff $H \subseteq X$ is closed it follows that G/H is indeed Hausdorff. □

Together with the fact that a quotient topology assumes 2nd countability of a topology, we get,

Lemma 4.3. *If $W = L(H)$, $W \oplus W' = L(G)$ and $p : G \to G/H$ projection, then $\psi = p \circ \exp|_U$ is a bijection for small enough neighborhood $0 \in U \subseteq W'$.*

Proof. Recall that exp is a diffeomorphism for small enough neighborhood of 0. It suffice to show that ψ is injective (why?), let $w_1, w_2 \in U$, and assume $\psi(w_1) = \psi(w_2)$, if $w_1 = -w_2$ then $\exp(w_1)H = \exp(-w_1)H \implies \exp(w_1)^2 H = H \implies 2w_1 \in W \implies w_1 = 0$. Assume otherwise $w_1 \neq -w_2$, $\exp(w_1)H = \exp(w_2)H \implies \exp(w_1 - w_2) \in H$, then $w_1 - w_2 \in W \implies w_1 - w_2 = 0$. Therefore ψ is a bijection composited by continuous functions, therefore continuous. □

Exercise 4.1. Show that ψ is a homeomorphism, in other words, show that its inverse is continuous.

Proposition 4.4. *G/H can be given a smooth manifold structure, such that $p : G \to G/H$ is smooth and any $f : G/H \to M$ is smooth if and only if there exists $F : G \to M$ smooth with $f = F \circ p$.*

Proof. Split $L(G) = W \oplus W'$ as before, such that $W = L(H)$ and let $0 \in U \subseteq W'$ be a small enough neighborhood as following the last lemma. Define the map $\psi : U \to G/H$ by $\psi = p \circ \exp|_U$, it follows from the lemma and exercise that it is a homeomorphism between U and a neighborhood of eH in G/H. In a similar matter define $\psi_x : U_x \to V_x$ by $\psi_v = \rho_x \circ \psi$, by the fact that $T_eG \cong T_xG$ it is a homeomorphism as well. $\psi_x^{-1}\psi : U \to U_x$ is smooth, it is clear by restriction of charts of G in e and x.

The second part is similar and is left as an exercise. □

Definition 4.5 (Bundle)**.** A **bundle** is structure consists of:

1. Bundle space: a topological space E, also called total space
2. Base space: another topological space, denoted usually by B
3. Projection: a continuous map $p : E \to B$ surjective

For $b \in B$ we call $F_b := p^{-1}(b)$ the **fiber** of b in the bundle.
A bundle is called a **fiber bundle** if also:

4. Fiber: There exists another topological space F such that F_b is homeomorphic to F for all $b \in B$
5. For any $b \in B$ there is an open neighborhood $b \in U \subseteq B$ such that there is a homeomorphism $\varphi : U \times F \to p^{-1}(U)$

Notation 4.6. We will denote a fiber bundle by $F \to E \to B$ to describe both the projection and the embedding of the fiber in the bundle space.

The concept of a bundle is not new to us, as any topological covering space is a fiber bundle such that F_b is discrete for any $b \in B$. In general, we wish to conceptualize a fiber bundle as a more general notion of a covering space, locally E assumes the structure of $U \times F$, meaning that when inspecting a small enough neighborhood, E can be thought of as a stack of U copies of the fiber.

Example 4.1. If M has a structure of a manifold (and in particular a smooth manifold) then $TM = \bigsqcup_{x \in M} T_x M$, the tangent bundle, is a bundle. We can choose $p : TM \to M$ by $p(x, u) = x$, it also follows that $F_x \simeq T_x M$, the tangent space of x.

Recall that if G assumes a Lie group structure then $T_x G \cong L(G)$, in other words $L(G) \to G \times L(G) \cong TG \to G$ is a fiber bundle.

Proposition 4.7. *$H \to G \to G/H$ defines a fiber bundle.*

A direct result is that for each $xH \in G/H$ there is an open subset $xH \in U \subseteq G/H$ can be identified as $U \times H$.

Exercise 4.2. Show that if $\dim G = n$ and $\dim H = k < n$ then $\dim G/H = n - k$.